

Natural Language Processing – CS6120

Assignment 2

Total Mark: 100%

Syntactic and Semantic Parsing

This assignment will test your learnings from Syntactic and Semantic Parsing lectures. Most of the assignment requires writing. You can either solve your assignment with pen and paper and submit a digital scan (pdf) OR create a digital version.

Estimated completion time: ~2 hours

Question 1: Syntactic Parsing (20%)

1.1 Consider the sentence: "*She announced a program to promote safety in trucks and vans.*" Draw the syntactic trees for at least three other interpretations for this sentence.

Hint: sample interpretations:

1. The program promotes safety in trucks and also promotes vans.
2. The program promotes safety in trucks and in vans.
3. The program promotes safety, and the program is in trucks and vans.

1.2 Draw the trees for at least two interpretations for each of the following sentences:

- *They are hunting lions and tigers.*
- *Monty flies like mosquitoes.*

You can use your own or the following grammar:

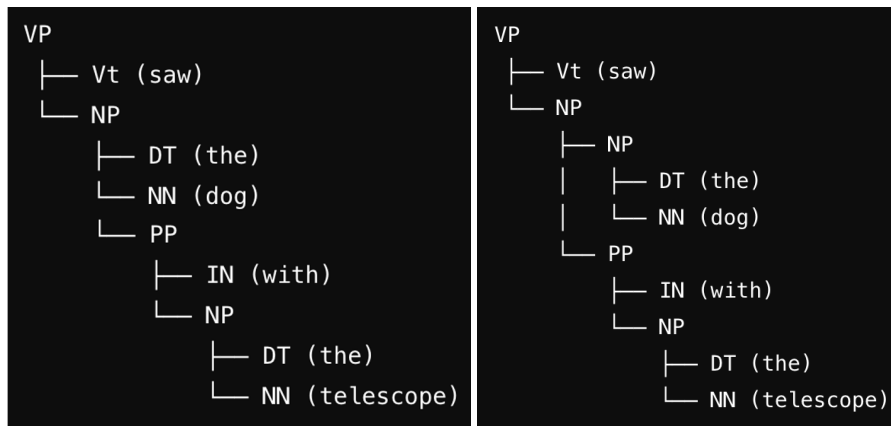
- **Grammar:**
 - $S \rightarrow VP$
 - $S \rightarrow NP VP$
 - $NP \rightarrow \text{Pronoun}$
 - $NP \rightarrow N$

- NP → Det N
- NP → Adj N
- NP → NP NP
- NP → NP PP
- NP → NP Conj NP
- VP → V
- VP → V NP
- VP → V PP
- VP → Aux VP
- VP → V NP PP
- VP → V Adj
- VP → VP Conj VP
- Aux → is | are | was | were
- Adj → hunting
- PP → P NP
- PP → PP Conj PP
- X → X Conj X (where X can be NP, VP, PP, Adj, N)
- Lexical:
 - Pronoun → They
 - Det → a
 - N → Monty | lions | tigers | flies | mosquitoes
 - V → announced | promote | hunting | flies | like
 - Adj → hunting
 - P → in | like | to
 - Conj → and
 - N/V → flies (can be a noun or a verb)
 - P/V → like (can be a preposition or a verb)

- **Adj/V → hunting** (can be an adjective or a verb)

Question 2: Precision and Recall in Parse Trees (10%)

Given the phrase "saw the dog with the telescope" and the gold parse tree (left) and the predicted parse tree (right):



Calculate the precision and recall of the predicted parse tree.

Question 3: Context-Free Grammar (CFG) Parsing (20%)

Consider the following CFG:

$S \rightarrow NP VP$

$NP \rightarrow DT NN \mid NP PP$

$PP \rightarrow IN NP$

$VP \rightarrow VB NP \mid VP PP$

$DT \rightarrow \text{the}$

$NN \rightarrow \text{man} \mid \text{dog} \mid \text{park} \mid \text{cat}$

$VB \rightarrow \text{saw}$

$IN \rightarrow \text{with} \mid \text{in}$

3.1 How many parse trees are there for the sentence "the man saw the dog in the park"?

3.2 How many parse trees are there for the sentence "*the man saw the dog in the park with the cat*"?

3.3 Consider a sentence that is grammatical under the above CFG and has exactly k prepositions following the verb and 0 prepositions before the verb. How many parse trees will this sentence have? Explain your reasoning.

Question 4: Chomsky Normal Form (CNF) (15%)

Given Grammar in Chomsky Normal Form (CNF):

$S \rightarrow AB$

$A \rightarrow BC$

$B \rightarrow CC$

$C \rightarrow AB$

$D \rightarrow a$

$E \rightarrow b$

$B \rightarrow b$

$C \rightarrow a$

Sentence: "*aabbb*"

Apply the CYK algorithm to parse the given sentence. Fill in the CYK parsing table and determine if the sentence can be generated by the grammar.

Question 5: Probabilistic Context-Free Grammar (PCFG) (15%)

Given PCFG:

$S \rightarrow NP VP [1.0]$

$NP \rightarrow Det N [0.5] \mid NP PP [0.5]$

$VP \rightarrow V NP [0.5] \mid VP PP [0.5]$

$PP \rightarrow P NP [1.0]$

$Det \rightarrow the [0.5] \mid a [0.5]$

$N \rightarrow cat [0.5] \mid dog [0.5]$

$V \rightarrow chased [1.0]$

P → with [1.0]

Sentence: *"the dog chased a cat with the dog"*

- 5.1 Draw all possible parse trees for the sentence using the given PCFG.
- 5.2 Compute the probability of each parse tree.
- 5.3 Identify the most probable parse tree.

Question 6: Transition-Based Dependency Parsing (20%)

Given Sentence: *"I shot an elephant in my pajamas."*

Dependency Relations:

I → shot (nsubj)

shot → elephant (dobj)

elephant → an (det)

elephant → in (nmod)

in → pajamas (pobj)

pajamas → my (poss)

Perform the transition-based dependency parsing using the shift reduce algorithm. Show each step (shift, left-arc, right-arc) until the final dependency tree is obtained.

Deliverables

- A (pdf) report containing your solution for each exercise. Please use the provided numbering to refer to each exercise.